

What is Shell Programming?

What is shell?

- Shell can be considered to be a special user interface program providing interface for using operating system services.
- Basically it is a command language interpreter.
- Shell
 - Command line shell: uses a command line interface (Terminal)
 - Graphical shell: uses graphical user interface (GUI)

What does shell look like?

- Shells are either interactive (accept commands from user and executes them)
- Or bunch of commands as input from a file (shell scripts or shell programs)
- A shell script is a text file with Linux commands.
- Shell script is saved with extension .sh
- Syntax of shell program or shell script is like any other programming language.

Why write shell scripts?

- To avoid repetition
 - For doing sequence of steps with standard commands over and over, use just one.
- To automate difficult tasks
 - Many commands have subtle and difficult options that users may not remember or find hard to use
- Routine backup
- Monitoring tasks
- Adding new functionality to shell!
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Why not write shell scripts?

- Prone to errors
- Slow execution
- May not be well suited for large and complex task!
- Minimal data structure

Example Shells

- Any shell provides the same functionality as the other. But the commands will be different and built-in functions will be different.
- BASH (Bourne Again SHell)
 - Most popular in Linux
 - Used as default login shell in Linux and MacOS
- CSH (CSHell)
 - Syntax and usage of CSH are similar to C programming language
- KSH (Korn Shell)
 - Was the base for POSIX shell standard specifications
- TCSH (Terminal based CSHell)
 - Enhanced version of Berkley Unix C Shell CSH

Shell programming overview

- Bourne shell variables are different from variables in csh and other shells!
- Operators
- Control statements
- Conditional statements
- Command line arguments (positional parameters)

Executing a shell script

- Shell program is created using text editor.
- However, this file is actually a program that will be executed.
- Hence, the mode of the file has to be made “executable”.
- Use “chmod” command to achieve the above.

Or like this..

- Create a file using a vi editor(or any other editor). Name script file with extension .sh
- Start the script with `#!/bin/sh`
- Write some code.
- Save the script file as filename.sh
- For executing the script type
 - `bash filename.sh`

Example shell scripting commands

- Basic Shell Scripting Commands in Linux:
 - cat, more, less, head, tail, mkdir, cp, mv, rm, touch, grep, sort, wc, cut

Example:

- **#!/bin/sh**
- `echo "Who are you? CAN YOU IDENTIFY YOURSELF?"`
- `read YOU`
- `echo "Hello, $YOU"`

Now... Its upto YOU!